Lecture Workshop on "Quantum Statistics-Theory and Experiment"

Organized by

St. Philomena's College (Autonomous), Mysore-Bangalore Road
Bannimantap, Mysuru - 570 015

Sponsored by

Indian Academy of Sciences, Bengaluru
Indian National Science Academy, New Delhi
The National Academy of Sciences, India, Ahmedabad

A lecture workshop on ***"Quantum Statistics-Theory and Experiment"*** will be held at St. Philomena's College (Autonomous), Mysuru during ***15-17 February, 2019***.
A brief outline of the course is as follows:

There will be 12 Lectures on the following topics:

1. Fermi - Dirac and Bose - Einstein Statistics
2. Consequences of Fermi - Dirac Statistics on the Transport and Magnetic properties of transition metals and Anisotropic and Giant Magnetoresistance.
3. Video demonstration of some experiments
4. Experiments on Bose - Einstein condensates
5. Nobel Prize in Physics - 2018 (Special Lecture)

Resource persons:

1. Prof. R. Srinivasan, Retired Director, UGC-DAE Consortium for Scientific Research, Indore (MP).
2. Prof. G. Srinivasan, Raman Research Institute, Bengaluru, (Retired).
3. Prof. T.G. Ramesh, Retired Head, Materials Science Division, National Aerospace Laboratories (NAL), Bengaluru.

Convener: Prof. R.Srinivasan

Co-ordinator: Prof. D. Revannasiddaiah,
Department of Physics, St. Philomena's College, Mysuru.
Email-id: **revannasiddaiah.d@gmail.com** , Cell : 9886576762

This workshop is open to M.Sc. Students (and also for some highly motivated B.Sc. Students), Research Scholars, University and College Teachers. Interested may register their names with the workshop Co-ordinator Prof. D. Revannasiddaiah.

Necessary instructions have been provided in the enclosed enrolment form. Our College web page http://www.stphilos.ac.in/ too has a copy of this form.

Last date for receipt of filled enrolment/registration form: February 10, 2019

Prof. D. Revannasiddaiah
Co-ordinator

St. Philomena's College(Autonomous)
Department of Physics

stphilos.ac.in
(+91) 821 4240 900
Bannimantap, Mysuru
570015, Karnataka, India

Enrolment form

Lecture workshop on
"Quantum Statistics-Theory and Experiment"

15–17 February, 2019

Applicant

Full name: _____

Designation: _____

E-mail address: _____

Mobile number: _____

Affiliation

Institution: _____

Address: _____

Signature of the Applicant

Name and Signature of the
Head of the Department/Institution

Instructions for participants:

1. Scan, fill and e-mail the enrolment form to **pgphysics.philos@gmail.com** or **revannasiddaiah.d@gmail.com** .
2. Alternatively, you may mail it to the following address: **The Head, PG Department of Physics, St. Philomena's College, Bannimantap, Mysuru -570 015**
3. No TA, DA and accommodation will be provided for the outstation participants (if any).
4. Last date for the submission of filled enrolment/registration forms: **February 10, 2019.**

SCIENCE ACADEMIES' THREE DAY LECTURE WORKSHOP ON

QUANTUM STATISTICS – THEORY AND EXPERIMENT

Department of Physics, St. Philomena's College, Mysuru
15-17 February, 2019

PROGRAM

Inauguration: Friday 15th February : 9 to 9-30 AM

Day and date	9-30 to 11 Hrs	11-30 to 13 Hrs	14-15-30 Hrs	16 to 17-30 Hrs
Friday 15 Feb 2019	GS1	GS2	TGR1	Special Lecture(SPL) RS
Saturday 16 Feb 2019	GS3	GS4	RS1	TGR2
Sunday 17 Feb 2019	RS2	TGR3	TGR4	RS3

Valedictory: Sunday 17th February: 17-30 to 17-45 Hrs

Tea Break: 11 to 11-30 Hrs , 15-30 to 16 Hrs ***Lunch Break: 13 to 14 Hrs***

List of lectures:

Dr. G. Srinivasan (GS)	GS1	Fermi-Dirac Statistics 1
	GS2	Fermi-Dirac Statistics 2
	GS3	Bose-Einstein Statistics 1
	GS4	Bose-Einstein Statistics 2

Dr. T.G. Ramesh (TGR)	TGR1	Transport properties of transition metals
	TGR2	Magnetic properties of transition metals
	TGR3	Anisotropic and Giant Magnetoresistance
	TGR4	Video demonstration of some experiments

Dr. R. Srinivasan (RS)	RS1	Flow properties of LHeII Second sound in LHe II
	RS2	Quantization of circulation super-fluids
	RS3	BEC in alkali metal vapours and verification of theory
	SPL	Nobel Prize in Physics - 2018

Venue: UG Auditorium,
St. Philomena's College (Autonomous),
Mysuru